C02 Camera Setup

CRE132 worksheet

What you'll make

A camera for a 2D pixel-art game. It keeps the art crisp at any window size, follows the player smoothly, and never shows past the edge of the world.

You'll set the camera's **orthographic size**, add a **Pixel Perfect Camera**, and use **Cinemachine** to follow the player with a dead zone and damping, and a **confiner** that keeps it inside the level. There's still no code: there's no movement script until C04, so you'll move the player by dragging it in the Scene view while the game runs.



Figure 1: The finished scene: a grass field bigger than the screen, ringed with trees, with the camera following the player.

Before you start

1. Unzip `CRE132_C02_CameraSetup_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_C02_CameraSetup_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Core/C02_CameraSetup`.
3. Put the **Scene** and **Game** views side by side: drag the **Game** tab to the right-hand edge of the Scene view. Press **Play**, select the **Player** in the Hierarchy and drag it around in the Scene view. The camera stays where it is, so you can drag the player right out of the Game view. Press Play again to stop.

The scene has a world bigger than the screen: **Ground** (a grass field 40 × 24 units, made as in C01), **Scenery** (trees round the edge and a few bushes and rocks inside), the **Player** in the middle, a **Global Light 2D** and a plain **Main Camera**. You'll turn that camera into one that follows the player.

1. Orthographic size

Select **Main Camera**. Its **Projection** is **Orthographic**: no perspective, so things don't shrink with distance, which is what you want for 2D.

Size is *half* the height of the view in world units. It is **5** at the moment, so the camera shows 10 units from top to bottom. Set it to **5.625**: the camera then shows 11.25 units from top to bottom. At 16 pixels per unit that is 180 art pixels tall. Try a few other sizes and watch the Game view zoom in and out, then set it back to 5.625.

2. Pixel Perfect Camera

Pixel art looks wrong when one art pixel covers 3 screen pixels and its neighbour covers 4. The **Pixel Perfect Camera** stops that by only scaling the art by whole numbers.

Add it: on **Main Camera**, **Add Component > Pixel Perfect Camera**, then set:

- **Assets Pixels Per Unit:** 16 (the same as the sprites' import setting)
- **Reference Resolution:** X **320**, Y **180** (the size of your game in art pixels; 180 matches the camera size above)
- **Grid Snapping: Pixel Snapping**

Test it: in the Game view, pick different resolutions from the drop-down at the top (for example 1280×720). The art stays crisp: at 1280×720 every art pixel is exactly 4×4 screen pixels. Try **Grid Snapping: Upscale Render Texture** to see the true low-resolution look, then set it back to **Pixel Snapping**.

3. A Cinemachine follow camera

Cinemachine is a Unity package for camera behaviour. It's already installed in this project: open **Window > Package Manager**, choose **In Project** and you'll find **Cinemachine** in the list. (In your own projects, install it from **Unity Registry**.)

Add it: choose **GameObject > Cinemachine > Targeted Cameras > 2D Camera**. Rename the new object **CM Follow Camera**. Then select **Main Camera**: Unity has added a **Cinemachine Brain** to it.

- **Only the Main Camera renders.** A Cinemachine Camera is a set of instructions: it decides *where* to look. The Brain on the Main Camera moves the real camera there.
- The Cinemachine Camera now controls the lens too, so on **CM Follow Camera** set **Lens > Orthographic Size** to **5.625**.

Follow the player: on **CM Follow Camera**, drag the **Player** from the Hierarchy into **Tracking Target**.

Test it: press Play and drag the player around in the Scene view. The camera follows.

4. Dead zone and damping

The 2D Camera came with a **Position Composer** component. On it, set:

- **Dead Zone:** tick **Enabled**, **Size 0.3** × **0.3**
- **Damping:** X **1**, Y **1**

The dead zone is measured as a fraction of the screen: 0.3 is 30% of its width and height. While the player stays inside that middle area, the camera doesn't move, so small steps don't shake the view. With CM Follow Camera selected, the Game view draws guides: the clear area in the middle is the dead zone. **Damping** makes the camera catch up smoothly instead of snapping; higher numbers are slower.

Test it: play and drag the player in small steps, then in big ones.

5. Keeping the camera inside the level: Confiner 2D

Make the bounds: GameObject > Create Empty, rename it **Camera Bounds**, set its Position to 0, 0, 0. **Add Component > Polygon Collider 2D**. Tick **Is Trigger**, so it never blocks anything; it's only there to describe a shape.

Now make the shape cover the grass exactly. Click **Edit Collider** and drag the points onto the four corners of the grass (Ctrl-click a point to delete it; the default shape has five). The corners are at (-20, −12), (20, −12), (20, 12) and (-20, 12): you can also type them in under **Points**. Click **Edit Collider** again to finish.

Add the confiner: select **CM Follow Camera**. At the bottom of its Cinemachine Camera component, click **Add Extension** and choose **Cinemachine Confiner 2D**. Drag **Camera Bounds** into **Bounding Shape 2D**. Check that **Damping** is **0.5** (set it if not): it softens the stop at the edge.

Test it: play and drag the player into a corner. The camera stops at the edge of the grass, and the player walks on towards the edge of the screen.

6. Cinemachine Pixel Perfect

On **CM Follow Camera**, click **Add Extension** again and choose **Cinemachine Pixel Perfect**. Without it, the Cinemachine Camera's lens fights the Pixel Perfect Camera over the size of the view.

Test it: play, drag the player about and change the Game view's resolution. The art stays crisp and the camera still follows and stays inside the bounds.

Check your work

- ☐ Main Camera: Orthographic, Size 5.625, a **Pixel Perfect Camera** (16, 320 × 180, Pixel Snapping) and a **Cinemachine Brain**.
- ☐ CM Follow Camera follows the Player, with a dead zone of 0.3 × 0.3 and damping 1, 1.
- ☐ Small moves don't shift the camera; big moves are followed smoothly.
- ☐ The camera never shows past the edge of the grass.
- ☐ Camera Bounds: a **Polygon Collider 2D** with Is Trigger ticked, its four corners on the corners of the grass.
- ☐ Compare with `CRE132_C02_CameraSetup_Finished` if something behaves differently.

Key points

- **Only the Main Camera renders.** Cinemachine Cameras are instructions for it, carried out by the Cinemachine Brain.
- **Orthographic size is half the screen height** in world units.
- **Pixel perfect** means the pixels per unit, the reference resolution and the camera size all agree, so every art pixel covers the same number of screen pixels.
- **A confiner needs a collider shape**, not a solid wall: tick Is Trigger so it doesn't block anything.

If something goes wrong

- **The camera doesn't follow:** the **Tracking Target** slot on CM Follow Camera is empty.
- **There's no Cinemachine menu:** check that the Cinemachine package is in **Package Manager > In Project**.
- **The camera still shows past the grass:** the **Bounding Shape 2D** slot is empty, or the collider's points don't match the grass. If you edit the collider during Play, click **Invalidate Bounding Shape Cache** on the Confiner 2D (or stop and play again).
- **The art shimmers or looks blurry while the camera moves:** check the **Cinemachine Pixel Perfect** extension is on CM Follow Camera.

- **The view is the wrong size:** check the Lens Orthographic Size on CM Follow Camera, not only the Main Camera's Size.

Try it

1. Set the dead zone to 0, then to 0.8. Which feels better for a top-down game? For a platformer?
2. Turn damping off (0, 0). What changes?
3. Move one corner of the **Camera Bounds** collider inwards, so the shape isn't a rectangle. Where does the camera stop now?